

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	07 Jul 2026
Project Name	SHOPEZ E-commerce Application
Maximum Marks	4 Marks

Technical Architecture:

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Responsive web interface for customers to browse products, manage cart, wishlist, and orders	HTML5, CSS3, JavaScript, React.js, React Router
2.	Application Logic-1	User authentication, registration, login, and JWT token validation	Node.js, Express.js
3.	Application Logic-2	Product management, cart management, wishlist, and order processing	Node.js, Express.js
4.	Application Logic-3	REST API handling, business logic, and database communication	Express.js, Axios
5.	Database	Stores users, products, carts, wishlists, and orders	MongoDB (NoSQL)
6.	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7.	File Storage	Stores product images and static assets	Local File System / Public Folder
8.	External API-1	HTTP communication between frontend and backend	REST API using Axios
9.	External API-2	Authentication using JSON Web Tokens	JWT (jsonwebtoken)
10.	Infrastructure (Server / Cloud)	Frontend deployed on Vercel, Backend deployed on Render, Database hosted on MongoDB Atlas	Vercel, Render, MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and backend developed using open-source frameworks	React.js, Node.js, Express.js, MongoDB, Axios
2.	Security Implementations	Password hashing, JWT authentication, protected routes, environment variables for secrets	bcrypt.js, JWT, dotenv, CORS
3.	Scalable Architecture	Three-tier architecture consisting of Presentation Layer, Business Logic Layer, and Database Layer	React.js + Express.js + MongoDB
4.	Availability	Cloud deployment ensures high availability with managed database services	Vercel, Render, MongoDB Atlas
5.	Performance	Efficient REST APIs, React component rendering, optimized database queries, and fast API communication	React.js, Axios, MongoDB Indexing